AMME3060 ENGINEERING METHODS

Assignment #2 Due 30 October 11.59pm (Week 12, Friday);
This assignment is estimated to take 15 hours to complete. Weighting = 10% (oral)+ 3%(written submission)
This assignment is to be completed in groups of 3. There are individual and group marks. Groups are to be the same as for assignment 1 unless prior approval is given, and all prior group members wish to change group composition, and this agreement is reached by Tuesday of week 10.

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1 Overview of Requirements

You are an engineer working in a company that manufactures building products including double glazed windows, sky lights and solar water heaters. The sky lights are comprised of two sheets of poly-carbonate sheet in an insulated frame with the cavity interior filled with air. The water heaters are a water filled cavity with transparent glass on one side to allow radiative heating. In both products, the solar heating leads to buoyancy driven flow inside a rectangular cavity. The engineering team needs to determine the energy efficiency of these products under different conditions. Your task is to develop a custom numerical simulation tool which can be used to determine the heat distribution and heat transfer within the fluid in the fluid cavity. The flow within the fluid walls can be approximated as two dimensional and two standard configurations are shown in figure 1.

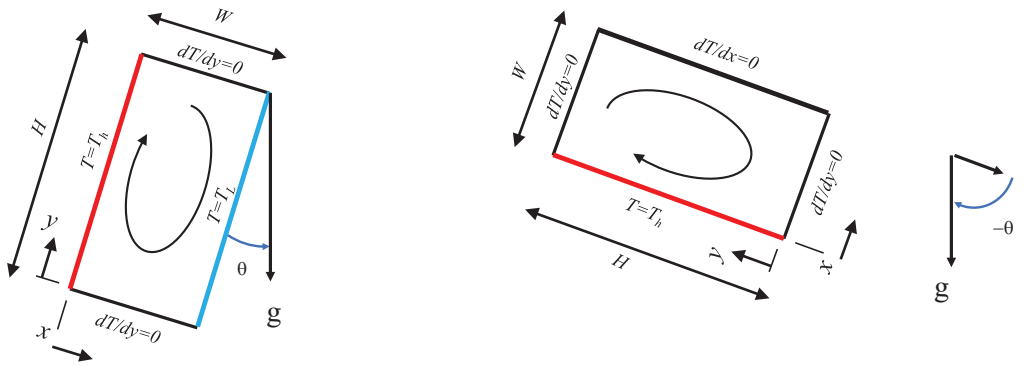


Figure 1: (a) Air filled skylight with heated and cooled boundaries; (b) Water filled solar water heater with heated lower wall and insulated top and side walls.

General Problem Description
The cavity is filled with a fluid. The cavity has a height of H and width of W and inclined at an angle of θ. The velocity of the fluid on all walls is zero. The initial velocity of the fluid in the x and y directions is $U(x, y) = 0$ and $V(x, y) = 0$ respectively for all cases. For all cases here, you can assume the fluid is incompressible. The fluid in the cavity has thermal diffusivity α and kinematic viscosity ν with values given below. Your task is to write a solver 'package' which will obtain accurate numerical simulations of the velocity and temperature in the cavity with time for the following two scenarios:

1. Case 1: Skylight Steady Solver: The working fluid is air ($\alpha = 22.0 \times 10^{-6}\text{m}^2/\text{s}$, $\nu = 1.557 \times 10^{-5}\text{m}^2/\text{s}$, $\beta = 0.00335\text{K}^{-1}$, $\kappa = 0.02597\text{W/mK}$). The walls at $x = 0$ and at $x = W$ are held at a constant fixed temperatures of $T = T_H$ and $T = T_L$ respectively. The $y = 0$ and $y = H$ walls are perfectly insulated. The initial temperature $T(x, y)$ of the fluid in the cavity is $(T_H + T_L)/2$ and $T_H = 25$ and $T_L = 20$, $H = 0.1\,\text{m}$, $W = 0.025\,\text{m}$ and $\theta = 45°$.

   The task is to develop a transient solver to obtain accurate solutions of the steady state temperature and velocity field.

2. Case 2: Water Heater Fast Transient Solver: The working fluid is water ($\alpha = 1.43 \times 10^{-7}\mathrm{m}^2/\mathrm{s}$, $\nu = 0.893 \times 10^{-6}\mathrm{m}^2/\mathrm{s}$, $\beta = 2.59 \times 10^{-4}\mathrm{K}^{-1}$, $\kappa = 0.607\mathrm{W/mK}$). $H = 0.40\,\mathrm{m}$, $W = 0.02\,\mathrm{m}$ and $\theta = -85°$. The $x = 0$ wall has a fixed temperature of $T_H = 30$. All other walls are insulated. The initial temperature $T(x, y)$ of the water in the cavity is $T_L = 20$.

   The task is to develop a transient solver to obtain solutions for the temperature field and report on the solution at time $t = 50\,\mathrm{s}$. This solver must be fast and provide 'accurate enough' solutions. The marking scheme will include a weighting rewarding fast solvers. A simulation calculation time of less than 300s on a PC lab computer should be possible. The accuracy requirement is that average temperature of the fluid in the domain is accurate to within $\pm 0.3°\mathrm{C}$ of the true solution. The solution should be physical and with numerical stability well controlled.

Detailed Modeling Requirements
A working Navier Stokes solver code has been provided. The numerical details of this solver are contained in §2 and §3. Your task is to complete the simulation tool by adding in additional equations and numerical methods needed to solve for the temperature i.e. the transient 2D energy equation. You do not need to modify the Navier-Stokes solver, which is fully implemented already.

The choice of the numerical method you use to implement your energy equation solver in each case is up to you. You may use different methods for each case if you wish. You must submit a separate Matlab script for each case. Further constraints include:

1. your code and method should be efficient, and for the fast solver there is a run time limit. Run time checks should be performed and reported on. Runs should be performed on an AMME PC Laboratory computer (and submitted as a standard Matlab script and NOT run with any special parallelisation).

2. you need to demonstrate the code can obtain accurate results and that numerical stability is well controlled.

3. you need to report on the heat transfer. Use the Nusselt number :

$$\mathrm{Nu}_L = \frac{qW}{\kappa(T_H - T_L)}$$

   and where q is the average heat transfer over the hot wall:

$$q = \frac{1}{H}\int_0^H -\kappa\frac{\partial T}{\partial x}dy$$

4. in both solvers, report the temperature at the top wall. For Case 1 make the location $x = 0.0125$, for case 2 make the location $x = 0.005$. For the accurate solver, give the steady state temperature. For case 1 give the solution at full development, for case 2, give the temperature at $T = 50\mathrm{s}$. In case 2, also report the average temperature of the fluid in the domain.

Testing, Verification requirements
Once you have completed the code you are required to perform a range of tests to show your code is correct and implements the equations you intend as per your report.

1. You should provide accurate solutions demonstrating accuracy. These tests should clearly demonstrate understanding of error and for example include grid size and time step tests and any others as relevant.

2. You should report on any stability restrictions and test these.

Validation requirements
You should obtain solutions of case 1 using ANSYS Fluent and use the results to compare with your solutions. Through this comparison you are to quantitatively validate your code. You need to select the numerical methods you will use and justify these choices in your report. For Case 2 you can either use your

now validated case 1 code OR ANSYS Fluent to obtain accurate benchmark solutions.

Presentation of Solutions

Once you have shown your code meets the requirements above you will need to present your flow solutions obtained using both ANSYS Fluent and your Matlab code. This will require you to present a range of plots and tables to qualitatively and quantitatively describe both the transient flow development for each case and also describe the final flow state for case 1. Use these plots to demonstrate your code and method meets the requirements listed above. You may choose to use a range of contour plots and line plots and other means of presenting the data.

Submission and Reporting requirements

In a report you will present your method(s) for both the Matlab implementation and also the Fluent simulations; you will need to present your results and testing, validation and verification method and results. You will submit the code and also participate in a short oral exam of your work.

Your Matlab scripts should be extensively commented and every variable clearly defined. The Matlab code will be run by the marker and should be easily executed with clear instructions provided on how to do so for each of the test cases. Separate scripts should be provided for each test case.

2 Overview of the Navier-Stokes solver provided

The Navier stokes equations comprise of the Momentum equations and the conservation of mass equation (continuity). For incompressible flow with constant fluid properties, these can be written as,

Momentum equations:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + g_x\beta T$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + g_y\beta T$$

Conservation of Mass

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

where ν is the kinematic viscosity, p is the pressure, ρ is the constant density, u and v are the velocity components in the x and y directions, g is gravity and β is the thermal expansion coefficient for the fluid and T is the fluid temperature.

Solution of these coupled equations is possible through many algorithms. One common method is to solve the equations in stream-function-vorticity form (see Section 7.2.4 of Ferziger, Peric and Street (2020)). The details of this method are beyond the scope of the course but in brief, it involves taking the curl of the momentum equations to eliminate pressure. If this is done, a new equation for the transport of vorticity is obtained. Instead of solving the three equations given above, we now can solve the three new equations given below:

Vorticity transport equation:

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + v\frac{\partial \omega}{\partial y} = \nu\left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2}\right) + g_y\beta\frac{\partial T}{\partial x} - g_x\beta\frac{\partial T}{\partial y} \tag{1}$$

Stream Function Equation:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega \tag{2}$$

Velocity Components:

$$\frac{\partial \psi}{\partial x} = v, \qquad \frac{\partial \psi}{\partial y} = -u \tag{3}$$

The algorithm which implements a numerical method to solve these equations is described briefly in §3.

In addition to the Navier Stokes Equations, the Energy Equation (equivalently called the heat equation or Temperature transport equation) must be solved to obtain the temperature field as it evolves in time. For incompressible flow this equation can be written as:

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right), \tag{4}$$

The energy equation has not been implemented in the code provided and this is your main task in this assignment.

3 Solver Algorithm

The Navier-Stokes equations in stream function vorticity form given in §2 have been discretised using second order accurate centered finite difference scheme and a first order backwards in time scheme. The non-linear momentum equations are solved iteratively at each time step. The algorithm below is completed implemented in NStokestemplate.m apart from step 2.1 which is not implemented:

Step 1. Initialise the u, v, ω, T fields

Step 2. Advance the solution forward from time n to $n+1$: Calculate T^{n+1}, u^{n+1} and v^{n+1}

2.1. **Solve the Energy Equation (Eqn 4) to obtain the temperature at the next time step : T^{n+1}.**

2.2. Solve the Navier-Stokes Equations to obtain u^{n+1} and v^{n+1}. Use ω^n, u^n and v^n as initial guess of values ω^{n+1}, u^{n+1} and v^{n+1} and iterate until convergence.

2.2.1 Apply boundary conditions for ω.

2.2.2 Solve Eqn 1 to obtain the new value of ω^{n+1} using the current value of u^{n+1}, v^{n+1} and T^{n+1}

2.2.3 Solve Eqn 2 to obtain the new value of ψ^{n+1} using the new value of ω^{n+1}

2.2.4 Solve Eqns 3 to obtain the new value of u^{n+1} and v^{n+1} using the new value of ψ^{n+1}

2.2.5 Test for difference between new and previous values of u^{n+1} and v^{n+1}

2.2.6 If difference is greater than specified tolerance goto 2.2.1

2.3. repeat from Step 2 until flow time = final time

The entire code above has been implemented in NStokestemplate.m apart from step: 2.1. **Your task is to discretise the energy equation (Eqn 4) in space and time and implement a numerical method to solve this equation**. The choice of the discretisation and numerical method is up to you, but it must consist of one of the methods covered in this unit of study and it must satisfy the requirements outlined in §1.

The code provided in NStokestemplate.m is briefly commented and has guidelines on where to put your main energy equation solver.

A reminder that the solution method and code given to solve the Navier Stokes equation using the stream-function vorticity method is outside the scope of AMME3060, i.e. you will not be assessed on it.

4 Suggestions

The assignment is a substantial amount of work that will test your understanding of the concepts across a broad section of the unit of study. The following points are suggestions to guide you through the assignment.

1. Read this assignment brief thoroughly and map out the tasks and deliverables.
2. There are two different flow scenarios that must be considered. Think about what method(s) would be suitable to solve these different flow types. There will be trade-offs to consider. Make a deliberate decision and record this in your report with your logic.
3. You could start your investigation by performing the ANSYS Fluent simulations to understand the flow features. You will need to think about what numerical methods to select in Fluent. After getting everything working, you can perform mesh dependence tests and other accuracy tests to ensure your benchmark solutions are accurate.
4. Once you have decided what method(s) to implement in your Matlab code, discretise the equation and write out the pseudo code for how to solve the equations including what variables to use.
5. Write the code, debug, test. If you have lots of bugs, consider writing a stand alone solver for the energy equation (i.e. not within the NStokes solver provided). This will make debugging much easier.
6. Once the code is working and giving sensible results, perform the accuracy and validation/verification tests. Keep a good record of all the settings you use as it is easy to forget or loose these numbers. Make sure your write out your solution to at least 9 significant figures when doing accuracy tests. (I normally write out to 14significant figures)
7. Write a report presenting your work as per §1 requirements. Submit the code with extensive commenting.
8. re-read the marking rubric and requirements listed in §1 before submission to make sure you haven't missed anything.

5 Submission

Checkpoint 1: Due 16 October 11.59PM
Upload your meeting minutes so far and any Matlab files you have started to generate.

Checkpoint 1: Due 23 October 11.59PM
Upload your meeting minutes for the week since the 16 October and any Matlab files and other documents you have started to generate.

Canvas Final Submission: Due 30 October 11.59PM
There is no fixed page limit but I suggest a concise 8-10 page presentation of your results and method addressing all of the requirements listed in §1. You can place all derivations in an appendix written by hand and scanned.

The assignment files must be submitted through the course Canvas Site. Text-based similarity detecting software (Turnitin) will be used to detect plagiarism.

You should submit four files in a zipped folder:

i) Your report as .pdf : *GroupNameReport.pdf*
ii) Your Matlab files sensibly named with group name forming part of the name. Submit separate scripts for each solver.
iii) All your meeting minutes since the 24 October.

These files should be placed in a folder called: Groupname and zipped to create: *Groupsname.zip*. The code will be run by the marker.

Important
The report, meeting minutes and group contract should be submitted as a pdf only (**Microsoft word documents will not be marked**).

The zipped folder must be compressed using zip. **Other compression systems (e.g. RAR) will not be accepted and you will receive 0 marks** for this assessment.

Oral Exam
The oral exam will be scheduled during week 13 and will be held in your assignment groups. You will be given individual questions and be awarded individual marks. The total length of the exam will be about 20min. The focus of the exam will be on your assignment work, what you have learned from this and your understanding of the underlying numerical methods used in your assignment and related methods covered in the unit of study.

Academic Honesty
A reminder that the use of contract cheating sites such as Chegg are forbidden. If any student engages in this cheating, both group members will be held accountable. This is a group project and you are expected to work closely together.

Use of Generative AI as a tool to improve is OK for Assignment 2. Anything you submit should be something you have written (with assistance) and fully understand. Assistance must be acknowledged in detail (down to the line numbers of your code). The primary method of assessment of assignment 2 is via Oral exam. You will be asked questions that test your understanding. The assignment itself is a learning tool.

Rubric: Assignment 2 Oral Exam (/10)

Criteria	FA	PA	CR	DI	HD
Modeling, numerical methods and Matlab implementation (/6)	The code is incomplete, does not run or contains significant errors. Authors are unable to explain the code and method in detail in the oral exam.	The code and methods implemented contain significant errors and are not demonstrated to satisfy the requirements listed. The authors are able to explain the code but there are some gaps in understanding evident from the oral exam.	Sound implementation of method(s) to obtain numerical solutions to the two cases. The implementation may not meet some of the requirements or contain minor errors. Some gaps in understanding are evident in the oral exam.	Numerical methods are implemented which meet all the requirements for the two cases, but the logical case for these choices is not strong or the methods chosen may not be optimal. Authors demonstrate good but not outstanding understanding of the code and methods during the oral exam.	Numerical methods are selected with sound logical reasoning for each of the two test cases. The implementation meets all of the requirements listed. Strong understanding of accuracy, efficiency and numerical stability is shown. The discrete equations are obtained correctly and the numerical methods are implemented correctly. Outstanding understanding of the code and the characteristics of the numerical methods is demonstrated in the oral exam. Students will be able to apply concepts from the assignment broadly beyond the scope of the problem presented. Students should be able to demonstrate understanding of how other relevant methods from AMME3060 could be applied to this problem and any considerations that must be taken into account for their implementation.
Code verification and solution accuracy testing (/2)	An unsatisfactory attempt. Testing is not performed or is insufficient.	Accuracy testing is performed by results presented are only partly sufficient to quantify error in the solution.	As for DI but the testing is either brief or presentation and interpretation are lacking.	As for HD but the tests performed are not complete or do not provide strong verification of the methods implemented.	Extensive testing of the code is performed for all test cases, these tests verify the implementation of the algorithm including expected accuracy characteristics. The numerical error in all solutions presented is well controlled and accuracy is demonstrated. Strong understanding of all sources of error is shown.
Validation with ANSYS Fluent (/2)	The ANSYS model contains significant errors and the responses to questions in the oral exam are incomplete and show significant knowledge gaps.	As for CR but validation of all two test cases is incomplete or not demonstrated clearly in the report. The responses to questions in the oral exam are show significant knowledge gaps.	As for DI but convincing validation of all two test cases is not provided. The responses to questions in the oral exam show good understanding in most areas.	As for HD but the use of Fluent to obtain solutions may be less sophisticated or deliberate and the numerical error in the benchmark solutions not sufficiently well controlled. The responses to questions in the oral exam show strong understanding in most areas.	The ANSYS model is correct and a deliberate approach to selecting the numerical methods is taken. The solutions obtained are sufficiently accurate to use as benchmark solutions and error is well controlled. The ANSYS solutions are used to quantitatively and qualitatively validate the Matlab model. A strong understanding of model, the choices made, meshing and numerical error is demonstrated in the oral exam.

Rubric: Assignment 2 Written Submission (/3)

Criteria	FA	PA	CR	DI	HD
Report and Group Work(/3)	An unsatisfactory attempt. Non submission or very incomplete submission.	Missing significant elements which are required and is presented to a lower standard than CR. Group member contributions are not clearly provided for all components and submission of the weekly checkpoint uploads is missing or incomplete.	A sound attempt. As for DI but the report is missing some elements required. The accuracy testing is either brief or presentation and interpretation in the report are lacking. Individual contributions are identified in detail and regular group meeting minutes and other files are submitted in each checkpoint.	The report is well presented and includes all the elements listed but which but provides less depth and rigor than for HD. Individual contributions are identified in detail and regular group meeting minutes and other files are submitted in each checkpoint.	An outstanding description of the investigation and methods used and which covers all the elements listed on §1 including extensive accuracy testing. The accuracy test are thoughtfully performed and very well presented in the report. The approach taken to modeling in both solvers is fully described. The ANSYS solution and numerical method (settings) are clearly and completely described in the report. The code is clearly commented and neatly structured. The code includes instructions for running. Individual contributions to the code writing are exhaustively provided in the code as comments. Individual contributions to the assignment are identified in detail and regular group meeting minutes and other files are submitted in each checkpoint.